Effective dimensions retain complete ownership

Example: d = 8 coordinates -> D = 3 exact K classes; K and V have independent partitions

a Exact classes encode values and complete membership

All eight original coordinates remain assigned

$\widehat{K}_j$: 2 | -1 | 2 | 0 | 2 | -1 | 0 | 0

q_i: q0 | q1 | q2 | q3 | q4 | q5 | q6 | q7

axis 0 | axis 1 | axis 2 | axis 3 | axis 4 | axis 5 | axis 6 | axis 7

C0 = {0, 2, 4} K = 2

C1 = {1, 5} K = -1

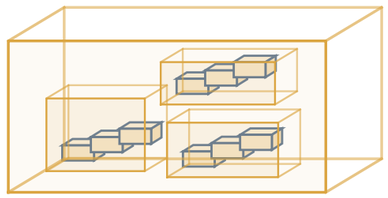
C2 = {3, 6, 7} K = 0

$\bar{q}_0 = q_0 + q_2 + q_4$ $\bar{q}_1 = q_1 + q_5$ $\bar{q}_2 = q_3 + q_6 + q_7$

$$s_{ij} = (2\bar{q}_0 - \bar{q}_1 + 0\bar{q}_2)/\sqrt{8}$$

Every different Q member is summed before multiplication.

Static 256-bucket BVH



Build once; reuse across layers and batches

Compact persistent domain

- Directory entry: (offset, D = 3)
- 3 descriptors: root / offset / length
- 3 FP32 representative values
- 8 member IDs, packed 4 per word

4 heads x 2 rows

Each Q keeps its own sums

Walk D descriptors

$\bar{q}_{i,a} = \sum_{d \in C_a^K} q_{i,d}$

$\sum_a K_a \bar{q}_{i,a}$

$$\bigcup_a C_a^K = \{0, \ldots, 7\}, \qquad C_a^K \cap C_b^K = \varnothing$$

The zero class is stored, even when its product is skipped.

b Independent V partition and the explicit member policy

5 | 5 | -2 | 5 | 1 | -2 | 1 | 1

0 1 2 3 4 5 6 7

V classes: {0,1,3} / {2,5} / {4,6,7}

V ownership differs from K ownership.

Default false

Optional true

5 | 0 | -2 | 0 | 1 | 0 | 0 | 0 — Representatives only

0 1 2 3 4 5 6 7

5 | 5 | -2 | 5 | 1 | -2 | 1 | 1 — Every member axis

0 1 2 3 4 5 6 7

Per history contribution: $w_{ij}V_{j,d}$; accumulate over j, then divide by Z_i.